

B. Tech Degree VIII Semester Examination, May 2006**ME 802 MACHINE DESIGN II***(2002 Admissions)*

Time : 3 Hours

Maximum Marks : 100

*(Design Data Book is permitted)**(Assume reasonable data wherever necessary)*

- I. (a) Explain how does a coupling differ from a clutch. (4)
 (b) Define - (i) self energizing brake (ii) self locking brake (4)
 (c) A band brake is to be designed for a winch to lift a load of 20KN through a 15m height by a rope wire wound on a barrel of 450mm diameter. The hoisting cycle is 3 minutes out of which actual braking time is 60 seconds. The angle of contact between the band and the brake drum is 210° . The brake drum may be keyed to the same shaft. Give the complete design. (17)
- OR**
- II. (a) Explain the chordal action in chains. (5)
 (b) Give the classification of chains. (3)
 (c) A V belt drive is to transmit 14.7 KW to a compressor. The motor speed is 1150 rpm and the compressor pulley runs at 400 rpm. Determine the size and number of belts required. (17)
- III. (a) Explain the term interference in involute profiles. (4)
 (b) What are the merits and demerits of helical gear? (4)
 (c) Design a worm gear speed reducer to transmit 22.5KW at 1440 rpm. The desired speed ratio is 24 : 1 and efficiency 85% . Assume that the worm is made of hardened steel and gear of phosphor bronze. (17)
- OR**
- IV. (a) State and explain law of gearing. (4)
 (b) Explain the nomenclature of helical gear with the help of a neat sketch. (4)
 (c) Determine the dimensions of a pair of 20° involute full depth spur gears to transmit 112.5KW, from a pinion running at 750 rpm to a gear running at 140 rpm. The service is intermittent with light shocks. (17)
- V. (a) Explain the hydro dynamic theory of lubrication. (5)
 (b) With a neat sketch explain the nomenclature of ball bearing. (5)
 (c) Design a journal bearing for a 10MW, 1000 rpm steam turbine which is supported by two bearings. Consider the bearing to be an average industrial bearing and operating temperature 60°C . (15)
- OR**
- VI. (a) Define the following terms :
 (i) Dynamic capacity (ii) Static capacity
 (iii) Cubic mean load (iv) Dynamic equivalent load (5)
 (b) What are the principal considerations while selecting a bearing. (5)
 (c) A shaft rotating at 1440 rpm is supported at 1440 rpm is supported by two bearings. The forces on each bearing are 6000N radial load and 3500N axial thrust. If shaft diameter is 40mm and the expected life of bearing is 500h. Select a suitable bearing if the require reliability of the bearing is to be 99% . (15)
- VII. (a) Explain various methods of representation of surface finish in working drawing. (5)
 (b) Explain the steps involved in the preparation of working drawings. (5)
 (c) What are the main design considerations for forged parts? (8)
 (d) What are the design considerations for machining? (7)
- OR**
- VIII. (a) How surface roughness is measured? (5)
 (b) What are the main design considerations for castings? (10)
 (c) Explain the design considerations for welded joints. (10)

